(a) one event on the material clock

hyperedge: Sn_{Ge} + V_{Ge} → ($Sn_{Ge}V_{Ge}$), window t = 3.2 ps

state before
$Ge_{0.88}Sn_{0.12}$ crystal,
2 vacancies

process step
MD anneal at 500 K,
0.5 ns

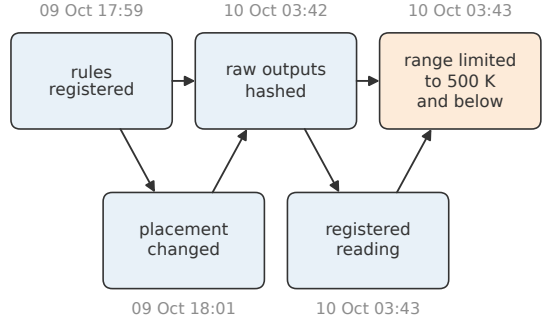
species before
Sn_{Ge}, V_{Ge}

balanced

species after
($Sn_{Ge}V_{Ge}$) pair

evidence source: MACE-MP-0 MD run for this study

(b) the same stage on the campaign clock



UTC time of each committed entry; an arrow means the later entry rests on the earlier one